

$$1,1 = 2$$

$$1,2 = 3$$

$$2,2 = 4$$

$$1,3 = 5$$

$$2,3 = 6$$

$$3,3 = 7$$

$$1,4 = 8$$

$$2,4$$

$$3,4$$

$$4,4$$

$$1,5$$

$$2,5$$

$$3,5$$

$$4,5$$

$$5,5$$

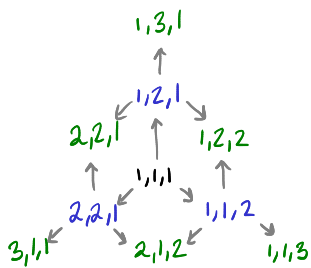
$$1,6$$

A hand-drawn multiplication table for the numbers 1 through 10. The numbers are arranged in a grid, with each row and column representing a number. The number 4 in the second row and second column is circled in blue.

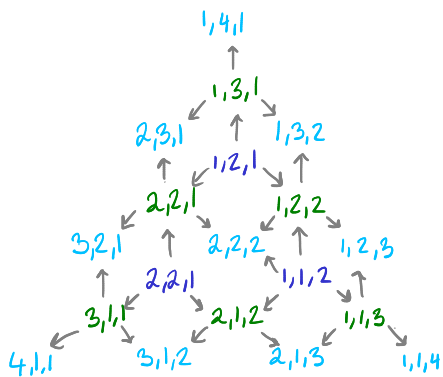
1	2	3	4	5	6	7	8	9	10
2	4	6	8	10	12	14	16	18	20
3	6	9	12	15	18	21	24	27	30
4	8	12	16	20	24	28	32	36	40
5	10	15	20	25	30	35	40	45	50
6	12	18	24	30	36	42	48	54	60
7	14	21	28	35	42	49	56	63	70
8	16	24	32	40	48	56	64	72	80
9	18	27	36	45	54	63	72	81	90
10	20	30	40	50	60	70	80	90	100



$$\Sigma = 4$$



$$\Sigma = 5$$



$$\Sigma = 6$$

3

1, 1, 1₁

4

1, 1, 2₂

5

1, 1, 3₃ 1, 2, 2₄

6

1, 1, 4₄ 1, 2, 3₆ 2, 2, 2₈

7

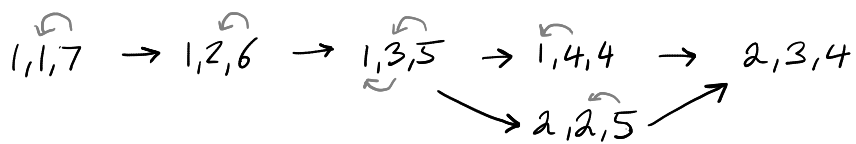
1, 1, 5₅ 1, 2, 4₈ 1, 3, 3₉ 2, 2, 3₁₂

8

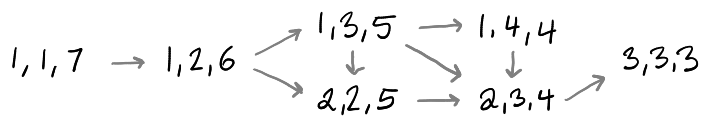
1, 1, 6₆ 1, 2, 5₁₀ 1, 3, 4₁₂ 2, 2, 4₁₆ 2, 3, 3₁₈

9

1, 1, 7₇ 1, 2, 6₁₂ 1, 3, 5₁₅ 1, 4, 4₁₆ 2, 2, 5₂₀ 2, 3, 4₂₄ 3, 3, 3₂₇



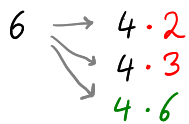
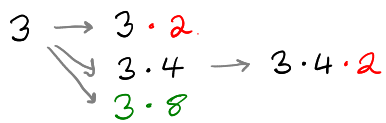
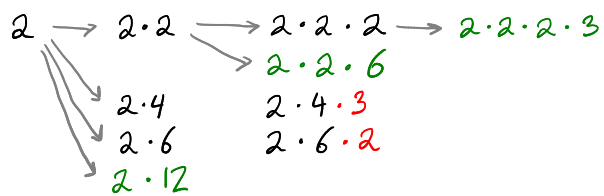
3,3,3?



1 → 1,1 → 1,1,7 7
1 → 1,2 → 1,2,6 12
1 → 1,3 → 1,3,5 15
1 → 1,4 → 1,4,4 16

2 → 2,2 → 2,2,5 20
2 → 2,3 → 2,3,4 24

3 → 3,3 → 3,3,3 27



$$12 \rightarrow 12 \cdot 2$$

$$9 \rightarrow 8 \cdot 3$$

$$6 \rightarrow 6 \cdot 4$$
$$6 \rightarrow 6 \cdot 2 \rightarrow 6 \cdot 2 \cdot 2$$

$$4 \rightarrow 4 \cdot 6$$
$$4 \rightarrow 4 \cdot 3 \rightarrow 4 \cdot 3 \cdot 2$$
$$4 \rightarrow 4 \cdot 2 \rightarrow 4 \cdot 2 \cdot 3$$

$$3 \rightarrow 3 \cdot 8$$
$$3 \rightarrow 3 \cdot 4$$
$$3 \rightarrow 3 \cdot 2 \rightarrow 3 \cdot 2 \cdot 4$$
$$3 \rightarrow 3 \cdot 2 \cdot 2 \rightarrow 3 \cdot 2 \cdot 2 \cdot 2$$

$$2 \rightarrow 2 \cdot 12$$
$$2 \rightarrow 2 \cdot 6$$
$$2 \rightarrow 2 \cdot 3$$
$$2 \rightarrow 2 \cdot 2 \rightarrow 2 \cdot 2 \cdot 2 \rightarrow 2 \cdot 2 \cdot 2 \cdot 3$$